

Class #8 - Matrix Exponential

1. If the characteristic polynomial of A does not have n _____ roots, then A may not have n linearly _____.

Example:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

has only two distinct eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 2$.

Consequently, the DE $\dot{x} = Ax$ has only _____ linearly independent solutions _____ and _____.

2. Problem: Suppose $n \times n$ matrix A has only $k < n$ linearly independent _____. Then, the DE $\dot{x} = Ax$ has only _____ linearly independent solutions of the form _____. How to find _____ linearly independent solutions?

3. Recall: The general solution to $\frac{dy}{dt} = ay$ is _____.

Analogously, we would like to say that $x(t) = e^{At}v$ is a solution of the vector DE

$$\dot{x} = Ax \quad (*)$$

for every _____ v . How do we define e^{At} though?

4. Def: For an $n \times n$ matrix A , the *matrix exponential*, denoted by e^{At} is defined as

$$e^{At} := I + At + \frac{A^2 t^2}{2!} + \cdots + \frac{A^n t^n}{n!} + \cdots.$$

5. It is known in Linear Algebra that the series converges for _____ t , and can be differentiated term-by-term:

$$\frac{d}{dt} e^{At} = A + A^2 t + \cdots = A(\text{_____}) = \text{_____}.$$

This implies that $e^{At}v$ is a solution of $(*)$ for every _____ v .

...

6. Remark: The matrix exponential e^{At} and the scalar exponential e^{at} satisfy many similar properties, for example,

$$(e^{At})^{-1} = e^{-At} \quad \text{and} \quad e^{A(t+s)} = e^{At}e^{As}.$$

However,

$$e^{At+Bt} = e^{At}e^{Bt} \quad \Longleftrightarrow \quad \underline{\hspace{2cm}}.$$

(Proofs are done in Linear Algebra.)

7. Back to #2:

Observe that

$$e^{At}v = e^{(A-\lambda I)t} \underline{\hspace{2cm}}$$

for any $\underline{\hspace{2cm}}$ λ since $\underline{\hspace{2cm}}$.

Moreover, $e^{At}v = e^{\lambda t}e^{(A-\lambda I)t}v$.

Suppose v satisfies $(A - \lambda I)^m v = 0$ for some integer m , then the infinite series $e^{(A-\lambda I)t}v$ terminates after $\underline{\hspace{2cm}}$ terms, and

$$e^{At}v = e^{\lambda t} \left(\underline{\hspace{4cm}} \right).$$

8. Algorithm for finding n linearly independent solutions of $\dot{x} = Ax$.

(i) Find all eigenvalues and eigenvectors of A . If A has n linearly independent eigenvectors:

(The infinite series $e^{(A-\lambda I)t}v$ terminates after _____term.)

(ii) Suppose A has only $k < n$ linearly independent eigenvectors (so, have k linearly independent solutions). To find additional solutions we pick an eigenvalue λ of A and find all vectors v such that $(A - \lambda I)^2v = 0$, but $(A - \lambda I)v \neq 0$. For each such vector v ,

$$e^{At}v = e^{\lambda t}e^{(A-\lambda I)t}v = e^{\lambda t}(v + \text{_____})$$

is an additional solution.

(iii) If we still do not have enough solutions, then we find all vectors v for which _____=0, but _____ $\neq 0$. For each such vector v ,

$$e^{At}v = e^{\lambda t}(v + \text{_____} + \text{_____})$$

is an additional solution.

(iv) We keep proceeding in this manner until we obtain n linearly independent solutions.